Modeling Discrimination with Causal Abstraction

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Abstract

A person is directly discriminated against only if their possession of a protected attribute, such as gender or race, causes their worse treatment. This implies that a protected attribute is sufficiently separable from other attributes to isolate its causal role. But gender is embedded in a nexus of social factors that resist isolated treatment. If gender (or another protected attribute) is socially constructed, in what sense can it cause worse treatment? Some suggest that since causal models require *independent manipulability*, i.e. the ability to isolate causal effects, attempts to causally model discrimination are misguided. Others suggest that we can preserve independent manipulability by saying that the perception of gender, rather than gender itself, causes worse treatment.

We address this problem differently. We introduce a framework for reasoning about discrimination, in which gender is a high-level *abstraction* of lower-level features. In this framework, gender can be modeled as itself causing worse treatment. Independent manipulability is ensured by allowing assumptions about how protected attributes are constituted to be precisely and explicitly stated, via an alignment between a protected attribute like gender and its constituents. Such assumptions can then be subjected to normative and empirical challenges, which lead to different views of when discrimination occurs. By distinguishing constitutive and causal relations, the abstraction framework pinpoints disagreements in the current literature on modeling discrimination, while preserving a precise causal account of discrimination.

Keywords: causation, discrimination, independent manipulability, social construction, causal abstraction

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1 Introduction

A person is directly discriminated against only if their possession of a protected attribute, such as gender or race, explains their worse treatment (Thomsen, 2018). When explanation is understood causally, we arrive at the notion of

> *Causal Discrimination:* A person is directly discriminated against only if their possession of a protected attribute, such as gender or race, causes their worse treatment.

The causal notion of discrimination is pervasive. It is invoked by U.S. discrimination law (Kohler-Hausmann, 2018), by leading algorithmic fairness criteria (Kilbertus et al., 2017; Kusner et al., 2017; Barocas et al., 2023; Plečko and Bareinboim, 2024; Loi et al., 2023), and by *audit studies*, in which experimenters detect institutional discrimination using techniques that resemble randomized controlled trials (Gaddis, 2018). For example, to test for gender discrimination in the job market, experimenters compare callback rates for resumes which are identical except for a gender-coded attribute, such as the applicant's name and pronouns (Galos and Coppock, 2023).

However, even assuming we have settled on a theory of which attributes should be protected from discrimination and why it is wrongful (e.g. Hellman 2008; Kolodny 2023), the causal notion of discrimination is difficult to spell out precisely. Indeed, on a counterfactual understanding of causation (Lewis, 1973; Woodward and Hitchcock, 2003a,b; Woodward, 2003), to determine whether gender caused worse treatment, we must consider whether an individual's treatment would have been worse, had their gender and *only* their gender been different.To take an extreme case, suppose we are testing for gender discrimination against Anne, a woman who was a Girl Scout, graduated from Barnard, became a nun, and is wearing a dress. How do we choose which of these attributes to vary, and which to hold fixed, when we manipulate gender? Because gender is bound up with so many other features of an individual, we lose a grip on the idea that it makes sense to intervene only on gender. A recent audit of racial discrimination concludes that interventions on race are similarly elusive: "Conceptually, it is challenging to rigorously define what it would even mean to manipulate an LLM's perception of a race in isolation from other factors" (Gaebler et al., 2024).

More generally, causal inference requires *independent manipulability*, i.e. that no intervention on a variable constitutes an intervention on other model variables. But one may wonder what exactly it means to intervene only on a person's gender, given that interventions on variables like names and self-presentation are plausibly partially constitutive of interventions on gender (Haslanger, 2017; Hu and Kohler-Hausmann, 2020). For instance, because names like 'Anne' form part of the social meaning of what it is to be gendered as a woman, an intervention setting a name of an applicant on a resume from Anne to Andrew cannot, in principle, be separated from an intervention on their gender. As a result, placing gender and names in a causal model together—as formal models of audit studies do—requires making a false independent manipulability assumption that names do not partially constitute gender. This leads to the worry that models of audit studies satisfy independent manipulability only if gender is treated as an implausibly "intrinsic" or "essential" feature of an individual, not constituted by names or any other model variables (Hu and Kohler-Hausmann 2020, pp. 4,

9; Hu 2024b, p. 9). We call this the *independent manipulability problem* for causal models of discrimination. We propose to reconstruct the problem as follows:

1. *Causal discrimination:* A person is directly discriminated against only if they are treated worse than others, and this is caused by their protected attribute.

2. *Independent manipulability:* Causal models require *independent manipulability*, i.e. that no intervention on a variable constitutes an intervention on other model variables.[1]

3. *Constituents in the Model:* The constituents of protected attributes are often manipulated or controlled-for by studies of discrimination, and thus appear in causal models of discrimination. For example, the model for an audit study intervening on names or appearance will include these as variables, and names and appearance plausibly partly constitute gender.

4. Therefore, causal models cannot express claims about discrimination: they would have to include protected attributes and their constituents, violating independent manipulability.

This worry has prompted many to defend causal models, typically by arguing that causal models of discrimination do not require independent manipulability for protected attributes such as gender or race, but rather for some distinct feature associated with those protected attributes. For example, Greiner and Rubin (2011) propose that racial discrimination occurs if others' *perception* of race causes worse treatment (cf. Singh and Wodak 2023), Sen and Wasow (2016) propose that race has several constituents and that race causes an outcome if any *constituent* causes that outcome, and Weinberger (2022) proposes that gender discrimination occurs if *signals* of gender cause worse treatment.[2] We argue that the problem is not adequately addressed by these proposals.

Those who developed the independent manipulability problem conclude that we must think of protected attributes like race or gender as explanatory in some sense that cannot be represented by traditional causal models (Kohler-Hausmann, 2018; Hu and Kohler-Hausmann, 2020; Dembroff et al., 2020; Hu, 2024b). Like the alternatives of skepticism (Holland, 1986) or agnosticism (Tolbert, 2024a) that race or gender is a cause, this is in tension with the notion of causal discrimination deployed by audit studies, algorithmic fairness criteria, the U.S. law, and those describing their experience of discrimination (Cherry and Bendick Jr., 2018, pp. 45-46, 53-55).

Crucially, all of the above responses to the independent manipulability problem share what we can call

[1] Woodward (2015) calls this requirement "independent fixability," Weslake (2024) calls it "independent manipulability," and Hu and Kohler-Hausmann (2020) call it "modularity." There are many related definitions; see Janzing and Schölkopf (2008) and Peters et al. (2017, pp. 17-21, 58-62).

[2] Bertrand and Mullainathan (2004) describe themselves as manipulating the interviewer's perception of race, rather than race itself (Bertrand and Mullainathan, 2004, p. 992). However, audit studies have also been seen as establishing direct, causal claims about protected attributes (Cherry and Bendick Jr., 2018). The issue of whether these studies manipulate protected attributes or the perception of them is orthogonal to the one we take up here, since the independent manipulability problem arises regardless (Hu and Kohler-Hausmann, 2024). Although our framework is technically agnostic between protected attributes and the perception of them, we focus here on the attributes. For discussion see §3.3.

> *the Single-Level Assumption:* When causally modeling some phenomenon, one must place all of the relevant variables in a single causal model, in which no two variables may describe the same facts at different levels of abstraction.

We jettison the Single-Level Assumption, by modeling gender as a high-level abstraction of lower-level attributes. We introduce

> *the Abstraction Model for Discrimination:* A person's gender is constituted by "low-level" natural and social attributes and relations. Gender causes an outcome if a change in gender, instantiated by a change in the low-level attributes constitutively aligned with gender, causes a change in outcome. This is the sense in which gender causes worse treatment, when gender discrimination occurs.

This model allows assumptions about how protected attributes are constituted to be precisely and explicitly stated, via an alignment between the attribute and its constituents. By distinguishing constitutive and causal relations, the abstraction framework dissolves the independent manipulability problem: in our model, gender and its constituents are not independently manipulable, in the sense that one cannot intervene on gender without intervening on its constituents. But as we explain, this is a benign (indeed intended) feature of abstraction, not an obstacle to causal modeling: when one attribute is simply a higher-level abstraction of other, lower-level attributes, one should expect changes at one level to correspond to changes at the other. While we develop the model by applying it to gender discrimination in audit studies, it applies equally to other protected attributes, such as race, disability, and religion, and in other settings, including algorithmic fairness and the U.S. law.

After introducing the abstraction model of discrimination (Section 2) and addressing the independent manipulability problem (Section 3), we apply our model to audit studies (Section 4). We introduce an initial model of audit studies (Section 4.1). We then consider several challenges for a successful model of audit studies. In particular, we consider the challenges that come with identifying which interventions are relevant to discrimination (Section 4.2), and we show how our framework allows us to explicitly model disagreements about how best to address these challenges when designing experiments that test for discrimination (Section 4.3). Finally, we turn to the worry that causal models cannot capture discrimination claims, because discrimination involves constitutive, rather than causal, explanation (Section 5). We argue that rather suggesting an alternative to the causal notion of discrimination, this worry points to an important distinction between two kinds of causal discrimination, which our framework has the expressive power to represent. In summary, we argue that a range of challenges in the literature provide reasons for objecting to particular causal models of discrimination. However, we argue that these are not objections to causal models of discrimination as such, because they can be stated within the framework of causal abstraction.

2 Abstraction and discrimination

The social construction of protected attributes like gender implies that an individual's gender is not a feature on par with other natural or social attributes and relations, but rather defined

in terms of such "lower-level" attributes and relations. While such multi-level representation is not part of a standard causal model, causal abstractions can represent causal systems at multiple levels of granularity. This section will introduce the framework of causal abstraction and then show how it can be used to model discrimination.

2.1 Introducing abstraction

Consider an experiment in which a bird is trained to peck at objects of any shade of red (Yablo, 1992; Woodward, 2021). When the bird pecks at some red object, say a crimson one, we can make two causal claims about it, at different levels of abstraction: that it pecks at the object because it is *crimson*, or that it pecks at an object because it is *red*. Each of these causal claims suggests an associated causal model, one in which perception of a fine-grained color (e.g., crimson, scarlet, cyan, turquoise) causes pecking behavior, and another in which perception of a coarse-grained color (e.g., red, blue) causes pecking behavior.

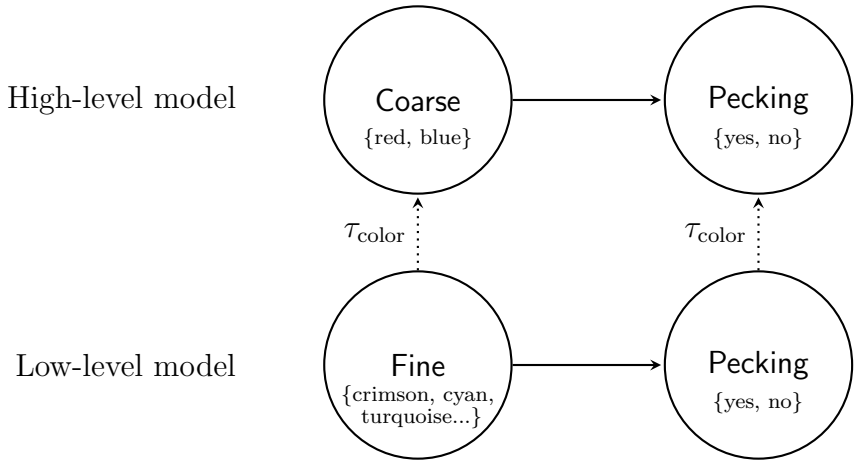


Figure 1: Color as an abstraction. An alignment τ_{color} maps values of the Fine color variable (crimson, scarlet) to values of the Coarse color variable (red). Solid arrows express causal relationships, e.g. that intervening on the value of the Coarse color variable can change the value of Pecking.

The fine-grained and coarse-grained models are depicted in Figure 1. The high-level model contains two variables: Coarse, which represents the bird's perception of a coarse-grained color, and Pecking, which represents whether the bird pecks. The causal arrow implies that Pecking is an effect of Coarse, and thus is some function of Coarse, perhaps together with random noise (included to capture variation among birds' pecking behavior not explained by object color). The low-level model is almost exactly the same, except it models pecking behavior as a function of fine-grained color Fine rather than coarse color.

Let us introduce the relevant notion of a causal model more formally.

Definition 1. A *structural causal model* is a tuple $\mathcal{M} = \langle \mathbf{V}, \mathbf{U}, \mathcal{F}, \mathbb{P} \rangle$, where $\mathbf{V}$ are the model variables, $\mathbf{U}$ are the noise variables, $\mathbb{P}$ is a distribution over $\mathbf{U}$, and $\mathcal{F}$ are the structural equations. The value of each noise variable $U \in \mathbf{U}$ belongs to $\text{Val}(U)$ and is determined by $\mathbb{P}(\mathbf{U})$. The value of each model variable $X \in \mathbf{V}$ belongs to $\text{Val}(X)$ and is determined by the

structural equation $f_X \in \mathcal{F}$ which is a function of the values of other variables:

$$X := f_X(\mathbf{V} \setminus \{X\}, \mathbf{U}).$$

Evaluating the structural equations according to the distribution of noise given by $\mathbb{P}$ extends it to an *observational distribution* over $\mathbf{V}$.

In other words, a causal model contains variables $X_1, X_2, X_3 \ldots$, where the value of each variable X is determined as a function f_X of the values of other variables, and perhaps some random noise, which captures variation not explained by model variables (see, e.g., Pearl 2000, Section 1.4.1; Peters et al. 2017, Definition 6.2; Bareinboim et al. 2022, Definition 1).

The effect of a manipulation on a variable is modeled as an *intervention* $\mathrm{do}(X = x)$, which replaces the function f_X defining X with a fixed value x (Pearl 2000, Section 1.4.3; Peters et al. 2017, Definition 6.8; Bareinboim et al. 2022, Definition 5). Because other variables may be defined as functions of X, the intervention $\mathrm{do}(X = x)$ percolates through the model, changing the likelihood that other variables take various values; this is the effect of the intervention. This gives us an *interventional* distribution on other model variables.

Definition 2. Given a causal model $\mathcal{M} = \langle \mathbf{V}, \mathbf{U}, \mathcal{F}, \mathbb{P} \rangle$, an *intervention* on a model variable $I = do(X_i = x_i)$ creates a new causal model $\mathcal{M}^I = \langle \mathbf{V}, \mathbf{U}, \mathcal{F}^I, \mathbb{P} \rangle$, which is identical to $\mathcal{M}$ except its structural equations, with $f^I_{X_i} = x_i$ and $f^I_{X_j} = f_{X_j}$ for $j \neq i$. The *interventional distribution* $\mathbb{P}^I$ denotes the joint distribution over $\mathbf{V}$ by extending $\mathbb{P}$ over $\mathbf{V}$ with $\mathcal{F}^I$. The definition can be applied to interventions on multiple variables,which replace multiple structural equations in $\mathcal{F}$ with constants.[3]

We are now in a position to explain the sense in which the high-level model "simplifies" the low-level one. Consider two models $\mathcal{M}_{\text{low}}$ and $\mathcal{M}_{\text{high}}$. An *alignment* τ is a function which maps values of variables in the low-level model $\mathcal{M}_{\text{low}}$ to values of variables in the high-level model $\mathcal{M}_{\text{high}}$ (Geiger et al. 2025, Definition 31; Rubenstein et al. 2017, Section 4.2; Beckers and Halpern 2019, τ in Definition 3.1 ff.):

Definition 3. An *alignment* τ between the causal models $\mathcal{M}_{\text{low}}$ and $\mathcal{M}_{\text{high}}$ maps values of variables $X_1, \ldots, X_k$ in $\mathcal{M}_{\text{low}}$ to values of variables $Y_1, \ldots, Y_\ell$ in $\mathcal{M}_{\text{high}}$:

$$\tau(X_1 = x_1, \ldots, X_k = x_k) = (Y_1 = y_1, \ldots Y_\ell = y_\ell).$$

Conceptually, an alignment may be thought of as a *simplification* of the low-level model. For example, it may be constructed by mapping different values of a low-level variable to individual values of that variable, mapping values over multiple low-level variables into values of a single variable, ignoring the values of some low-level variables entirely, or a combination of these operations (Geiger et al., 2025, Definitions 36-38). In the above example, the alignment τ_{color} merges the fine color values into the coarse color values by mapping crimson

[3]This is a mathematical definition. On certain theories of causality, to count as an intervention, a manipulation of a variable X, which seeks to reveal a possible effect on another variable Y, may need to satisfy several requirements besides independent manipulability. For example, according to Woodward and Hitchcock (2003a), the manipulation must be the only cause of X, must not directly cause Y (except via X), and must not share a common cause with Y (unless this cause affects Y via X).

and scarlet to red, while mapping cyan and turquoise to blue. As in this example, the alignment τ need not be injective, and as a result, interventions on high-level variables will often be ambiguous between several different interventions on low-level ones. (We discuss this issue further in Section 4).

So-defined, alignments are highly permissive: strictly speaking, any model can be aligned with any other. As we now explain, to define an *abstraction*, an alignment must also ensure that interventions in the low-level model and interventions in the high-level model are *causally consistent*. To check whether τ_{color} is causally consistent, we first perform an intervention on color at the low level, and then transform the effect it has on pecking to the high level via the alignment. We then check that this has the same result as first transforming the intervention via the alignment, and then intervening at the high level. This condition is in fact met by the alignment τ_{color}, because intervening on fine-grained color by changing it from crimson to cyan and then checking whether the bird pecks has the same result as intervening on coarse-grained color by changing it from red to blue and then checking whether the bird pecks. By mapping more general colors to corresponding more specific shades, τ_{color} tracks a real metaphysical relationship between a color and its many shades, to which pecking behavior is insensitive. More generally, an alignment τ between two models is causally consistent if intervening on variables in the low-level model $\mathcal{M}_{\text{low}}$ and then aligning into the high-level model $\mathcal{M}_{\text{high}}$ is the same as first aligning into the high-level model and intervening there (Rubenstein et al. 2017, Definition 3; Geiger et al. 2025, Definition 25):

Definition 4. An alignment τ between causal models $\mathcal{M}_{\text{low}}$ and $\mathcal{M}_{\text{high}}$ is *causally consistent* if intervening at the low level and then aligning to the high level is the same as first aligning the low-level intervention and then performing it at the high level. In other words, let $\mathbb{P}^{I}_{\text{low}}$ denote the interventional distribution over the values of all low-level variables that result from intervention I, let $\mathbb{P}^{\tau(I)}_{\text{high}}$ denote the interventional distribution over high-level values that results from intervention $\tau(I)$, and let $\tau(\mathbb{P}^{I}_{\text{low}})$ be the pushforward distribution, which transfers the distribution $\mathbb{P}^{I}_{\text{low}}$ onto high-level variables via τ.[4] Then causal consistency requires:

$$\tau(\mathbb{P}^{I}_{\text{low}}) = \mathbb{P}^{\tau(I)}_{\text{high}}.$$

Given such an alignment, we say that $\mathcal{M}_{\text{high}}$ is an *abstraction* of $\mathcal{M}_{\text{low}}$.

Causal consistency is a stringent requirement, requiring perfect correspondence between low-level and high-level interventions. We introduce it for simplicity, but it can be relaxed in various ways (see, e.g., Beckers et al. 2020; Rischel and Weichwald 2021; Geiger et al. 2025, Definition 41). Most simply, we can say that τ is *ϵ-causally consistent* when for some ϵ and all interventions I,

$$D(\tau(\mathbb{P}^{I}_{\text{low}}),\, \mathbb{P}^{\tau(I)}_{\text{high}}) < \epsilon,$$

where $D(\mathbb{P}_1, \mathbb{P}_2)$ is a distance measure between probability distributions, e.g. total variation (cf. Plečko and Bareinboim 2024, Appendix D1).

[4] Formally, $\tau(\mathbb{P}^{I}_{\text{low}})(\mathbf{y}) = \sum_{\mathbf{x} \in \tau^{-1}(\mathbf{y})} \mathbb{P}^{I}_{\text{low}}(\mathbf{x})$.

2.2 Modeling gender with an alignment

We now explain how to use an alignment to model the constitutive relations between protected attributes, such as gender, and their constituents. Any theory of what constitutes gender can be "plugged in" to the framework we present. However, we focus here on the view that gender is *constitutively socially constructed* (Mallon, 2024), in the sense that gender status is largely constituted by social facts, such as facts about social relations or social status (Haslanger, 2017, p. 20).[5]

The view that gender is socially constructed is attractive and plausible because it is thought to make good sense of transgender and genderqueer identities (Jenkins 2016; Dembroff et al. 2020, p. 4), as well as socially-determined variation in how intersex individuals are gendered (Ásta, 2013, pp. 726-7). However, our goal in this paper is not to provide a defense of social constructionism about gender. Instead, we are simply granting the view that gender is socially constructed to those who raise the independent manipulability problem (Hu and Kohler-Hausmann, 2020) and then arguing that our framework can accommodate this view and dissolve the problem.

In particular, we grant that the following attributes, which have been widely discussed in the literature on the independent manipulability problem, are constitutive of gender:

- Attributes of an individual's forms of address, such as their *name* or *pronouns* (Kohler-Hausmann and Dembroff, 2022, p. 90).
- Attributes of an individual's self-presentation, such as *wearing a skirt* or *makeup* (Hu, 2024a, p. 8).
- Attributes of an individual's education, such as *attending a women's college* (Hu, 2024a, cf. p. 11).

The idea is not that any change in one of these attributes is necessary or sufficient for a change in gender, but rather that some such changes, often together with others, constitute a change in gender.

The relations between gender and its constituents induce an alignment τ_{gen} that specifies how attributes constitute gender (Figure 2). To be sure, this already involves a substantive modeling assumption. One might hold that it is impossible, even in principle, to specify how exactly gender is constructed from lower-level attributes of individuals. For example, one might claim that gender, socioeconomic class, and race are all mutually constituted by each other, rather than by a more fine-grained basis of lower-level attributes.[6] However, we

[5]For social constructionist views of race, see for example Taylor (2022, pp. 139-144) and Mills (1998, pp. 51-55). While we focus on social constructionist theories of protected attributes, which throw the modularity problem into stark relief, causal abstraction can also be used to avoid modularity problems for non-social constructionist views. For example, some theories of race deny social constructionism (Spencer 2019, pp. 82-3, 95-6). On the simplest such theory, race is appearance and ancestry. Even on this view, your parent's races are partly constitutive of your race. Because some audits for fairness include both an individual's race and their parents' races in a single causal model (Kusner, Loftus, Russell, & Silva 2017), the independent manipulability problem arises even for such a simple, non-constructionist view of race, and causal abstraction provides a way to model the relationship between race and its constituents.

[6]For arguments that make room for this possibility by rejecting the traditional view that constitutive relations are asymmetric and irreflexive, see Barnes (2018) and Jenkins (2016), respectively. See also Bright et al. (2016) and Wang et al. (2022) for discussion of modeling intersectionality.

set this worry aside, because (1) this claim is potentially consistent with the abstraction model, insofar as we may model gender as abstracting race in one context, and model race as abstracting gender in another; and (2) we assume that social constructionist concerns regarding the possibility of causal models of discrimination do not rest on thorough-going skepticism that a social construction of gender in terms of lower-level attributes could ever be articulated in principle.

The simple assumption that gender aligns with these lower-level attributes already licenses answers to counterfactual questions about gender. For instance, it makes sense to ask what would happen were one to intervene on a person's gender in a specific way. This is just to ask what would happen, were one to intervene on the lower-level attributes of an individual in such a way that their gender would now be different, according to the constitution of gender, as defined by the alignment τ_{gen}. (We discuss whether this alignment is causally consistent, and thus defines an abstraction, in §4.)

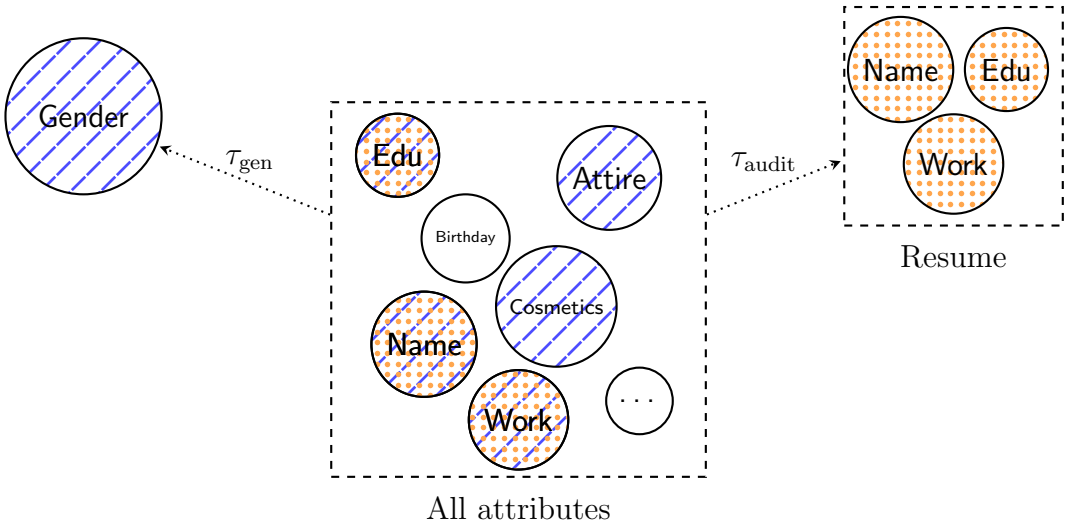


Figure 2: Toy example of gender and resume alignments. Abstracting over all attributes of an individual, the alignment τ_{gen} comes from a theory of how gender is constituted, and the alignment τ_{audit} are features listed on the resume used by audit studies. Some groups of attributes determine Gender (e.g., attire, pronouns), some groups of attributes determine gender and the resume (e.g. name, education), and others are discarded by both alignments (e.g. birthday).

3 The independent manipulability problem

In this section, we explain the requirement of independent manipulability and the problem it poses for causal models of discrimination. We then explain how the abstraction framework for modeling discrimination addresses this problem and compare it to some alternatives.

3.1 Introducing the problem

To ensure that interventions on individual variables reveal the effects of those variables, structural causal models must satisfy a requirement of independent manipulability (see, e.g.,

Pearl 2000, p. 63; Woodward 2003,p. 327; Peters et al. (2017), pp. 17-20, point 1). This requires that no intervention on any set of variables constitutes an intervention on variables outside that set. For example, a causal model which implies that object color causes bird pecking behavior satisfies independent manipulability because interventions on each of the two variables do not constitute interventions on the other. If we paint an object blue, we perform an intervention that directly changes just that object's color, not anything about the bird.[7]

Without independent manipulability, the causal effects of an intervention on a variable are hopelessly conflated with the intervention itself. Consider, for instance, a causal model claiming that the coarse color of an object causes its particular fine shade. This model violates independent manipulability because it is conceptually impossible to manipulate the color of an object separately from intervening on its particular shade (and often, vice versa). Since interventions on fine shades constitute interventions on coarse colors, we cannot isolate a causal effect of the intervention.

The independent manipulability problem, developed by Lily Hu and Issa Kohler-Hausmann (Kohler-Hausmann, 2018; Hu and Kohler-Hausmann, 2020), is that attributes like race, gender, and religion cannot be placed in a single causal model with the attributes from which they are socially constructed, because their constitutive relations violate the requirement of independent manipulability. In the case of audit studies, the problem is that gender and names cannot be placed in a single structural causal model without violating the independent manipulability assumption crucial to causal modeling and inference. The worry is that these two variables interact more like colors and their more specific shades than like bird pecking behavior and object colors; models including both variables inevitably violate independent manipulability.

The problem is easiest to see if we accept that gender is a social construction, and thus stands in *constitutive* rather than *causal* relations to many of its correlates. For then certain names constitute part of the social meaning of gender, and thus interventions switching names from "Anne" to "Andrew" constitute interventions to gender itself. This problem arises for any attributes that appear on a resume, like education or employment history, whenever they partly constitute gender, yet causal tests of discrimination often purport to hold these attributes fixed while manipulating gender.

3.2 Abstraction as a solution to the problem

The abstraction model of discrimination addresses the independent manipulability problem. Because protected attributes and their constituents are placed in *separate causal models*, and independent manipulability is only required within each model taken separately, this requirement is met in both the high-level model containing gender and the low-level model

[7]Of course, since variables often causally influence other variables, independent manipulability allows for an intervention on some variable to affect other variables through *causal* relations. Interventions on the color of the object plainly affect the bird's actual pecking causally, but color and pecking behavior are still independently manipulable. Instead, when variables X and Y are independently manipulable, this means that interventions on X do not constitute interventions on Y. While this distinction between causal and non-causal relations is an intuitive one, Janzing and Mejia (2022) discuss a number of puzzles about how to make it fully precise.

containing its constituents.

It is sometimes objected that structural causal models of discrimination cannot accommodate constitutive relations (Hu and Kohler-Hausmann, 2020, p. 5). This objection can be overcome with abstraction: the alignment nicely captures the relevant constitutive relations.

By dissolving the independent manipulability problem, the abstraction model immediately dissipates the concern that it is impossible to causally model gender in settings where its constituents are also endogenous (e.g., discrimination auditing). This in turn shows that independent manipulability does not in itself force causal models to posit an especially implausible metaphysical boundary between gender and the context in which it is socially embedded, e.g. by treating gender as an "intrinsic" or "essential" feature of individuals (Hu and Kohler-Hausmann 2020, pp. 4, 9, Hu 2024b, p. 9). Instead, the relationship between gender and other factors is modeled within the framework, via the alignment τ_{gen} that specifies how gender is socially constructed. This alignment may include all the attributes that are included in a particular social constructionist theory of gender. Although it excludes all constituents of gender, the high level model containing gender may also include variables representing causes and effects of gender.

We emphasize that the abstraction framework does not entail that gender, race, or other protected attributes are socially constructed. Instead, it provides a way of modeling the claim that these attributes are socially constructed, and in doing so addresses an independent manipulability worry that arises for social constructions. Though we find it plausible that at least some protected attributes are socially constructed, our argumentative strategy is to simply grant to those who raise the independent manipulability problem that protected attributes are socially constructed, and then to argue that this is not an obstacle to causally modeling them.

3.3 Comparison to other approaches

Our proposal develops Sen and Wasow's (2016) suggestion to treat social constructions (in their case, race) as a "bundle of sticks," such that manipulations on race are manipulations on some of the sticks. Our idea is to spell out this suggestion using an alignment τ_{gen} between two causal models, one with the constituents of gender and another with gender itself. We show in the next section how the framework allows us to distinguish several desiderata for a successful model of audit studies.

Concerns about protected attribute's manipulability lead Greiner and Rubin (2011) to propose that we use the perception of race, rather than race itself, as a treatment in causal studies of race. Some audit studies even explicitly frame themselves as intervening on perception (Bertrand and Mullainathan, 2004). But replacing protected attributes by their perception simply pushes the problem one step back: if race or gender is socially constructed, we should expect interventions on their perception to be bound up with interventions on the perception of names, education, and so on (cf. Hu and Kohler-Hausmann (2024)).Further, as we saw, U.S. discrimination law explicitly invokes the idea that attributes like sex or race themselves cause worse treatment—so there is value in seeing if this idea can be made to work, if want models of discrimination that can invoked in the courtroom.

Weinberger (2022) suggests that we can accommodate social constructionist theories of protected attributes by modeling audit studies as manipulating a signal of gender that is

causally downstream of gender. Audit studies vary names "as if" they were varying gender, thereby revealing an effect of gender along a particular causal path.[8] To include gender and names in the same causal model without violating independent manipulability, Weinberger makes two points. First, Weinberger proposes that in many narrow experimental contexts, even attributes that are constitutive of protected attributes send sufficiently "isolated" signals of these attributes, such that independent manipulability can be adopted as a limited modeling assumption, even if it strictly speaking false (Weinberger, 2022, p. 1282). Second, Weinberger observes that it does not follow from the assumption that a protected attribute is socially constructed that it is constituted by *all* of its signals (Weinberger, 2022, p. 1266). If gender is not constituted by some of its correlates, then, Weinberger suggests, it is possible to manipulate those correlates separately from gender.

We worry about both of these points. First, the stipulation that attributes like names, work, and education send isolated signals of protected attributes seems overly optimistic, given the empirical evidence (summarized by Hu 2024) that racial perception, even if induced just by a name, starts a cascade of racialized interpretation of other attributes. Second, if we avoid putative violations of independent manipulability by saying that the relevant feature is a signal (rather than a constituent) of the protected attribute, then we will need to model *every* feature that seems to violate independent manipulability as a signal (rather than a constituent) of gender. The model will need to include the signals we intervene on, as well as those we hold fixed. So not only names and pronouns, but also self-presentation, university, field of work, relations of subordination, and so on will need to be called signals (not constituents) of gender. When this is done, we seem to end up with nothing at all left in the constitutive basis. Whereas Weinberger explicitly frames his proposal as a way of accommodating social constructionism, his approach seems to rule out the view entirely, since this view treats many of the attributes that are varied or held fixed by audit studies as constituents of gender (Hu, 2024, §4; Hu and Kohler-Hausmann, 2024, pp. 249-51). By contrast, the abstraction model does not require us to reject the social constructionist views that motivate the independent manipulability problem: it allows that gender may be partly constituted by signals like names and pronouns.

4 Modeling discrimination

The abstraction model addresses the independent manipulability problem, but the use of abstraction does not obviate the need to make normative judgments about the relevant counterfactual contrasts in testing the effects of social kinds. Instead, the abstraction model provides a framework for making these judgments precise and explicit, primarily via an alignment between higher-level protected attributes and their lower-level constituents. In this way, the abstraction model offers the flexibility to encode different assumptions about how discrimination should be measured, which lead to different verdicts about whether discrimination has occurred. As we now show, once these assumptions are made precise and explicit, they can then be subjected to further normative and empirical challenges.

[8] VanderWeele and Robinson (2014) propose a model of race as a cause which is very similar to Weinberger's model. We focus on the latter because it is framed as a response to the independent manipulability problem.

We introduce an initial model of audit studies (Section 4.1) and distinguish several challenges for a successful model of audit studies (Section 4.2). In particular, we consider the worry that interventions which purport to test for discrimination in fact reveal quantities which are in some sense irrelevant to discrimination (Hu and Kohler-Hausmann, 2020). Stated precisely, this worry provides reasons for objecting to particular causal models of discrimination. However, we argue that it is not a problem for causal models of discrimination as such, because it can be stated within the framework of causal abstraction. Finally, we show how the abstraction framework has the expressive power to represent and clarify disagreements about whether discrimination occurred and how we should test for it (Section 4.3).

4.1 Modeling audit studies with abstraction

We can use the abstraction framework to propose a causal model of audit studies, by introducing two high-level models and alignments. Suppose some underlying causal model describes how low-level features of an applicant influence interview callback decisions. Audit studies select a strict subset of an individual's features, namely those appearing on their resume, "screening off" the effects of all other attributes of the individual on the outcome. In the terminology of abstraction, this defines an alignment τ_{audit}, which discards any attributes not appearing on the resume and trivially aligns all attributes appearing on the resume with themselves (Figure 2). The resume is assumed to include all attributes of an applicant that exert a causal effect on interview status.

Just for concreteness, consider a toy low-level model of interview callbacks where every individual has only six features: Attire, Birthday, Cosmetics, Edu, Name, Phone and Work. The alignment τ_{audit} is defined as follows:

$$\begin{aligned}&\tau_{\text{audit}}(\mathsf{Attire} = \mathrm{v_a}, \mathsf{Birthday} = \mathrm{v_b}, \mathsf{Cosmetics} = \mathrm{v_c}, \mathsf{Edu} = \mathrm{v_e}, \mathsf{Name} = \mathrm{v_n}, \mathsf{Phone} = \mathrm{v_p}, \mathsf{Work} = \mathrm{v_w}) \\ &\quad = (\mathsf{Edu} = \mathrm{v_e}, \mathsf{Name} = \mathrm{v_n}, \mathsf{Phone} = \mathrm{v_p}, \mathsf{Work} = \mathrm{v_w})\end{aligned}$$

In other words, τ_{audit} simply discards all information except for name, education, and work (i.e., attire, birthday, and cosmetics). In this way, τ_{audit} is by construction a "lossless" (and thus perfectly consistent) abstraction of "All attributes" when it comes to interview status.

A second alignment, τ_{gen}, describes how low-level features of an individual constitute gender. In addition to drawing from features on the resume, τ_{gen} may draw on attributes left out by τ_{audit}. Again for the sake of concreteness, suppose that Attire, Cosmetics, Education, Name, and Work each take one of two possible values, which are gendered masculine and feminine. The alignment τ_{gen} is defined as follows:[9]

$$\begin{aligned}&\tau_{\text{gen}}(\mathsf{Attire} = \mathrm{v_a}, \mathsf{Birthday} = \mathrm{v_b}, \mathsf{Cosmetics} = \mathrm{v_c}, \mathsf{Edu} = \mathrm{v_e}, \mathsf{Name} = \mathrm{v_n}, \mathsf{Work} = \mathrm{v_w}, \dots) \\ &\quad = \begin{cases} \text{woman} & \text{if a majority of } v_a, v_c, v_e, v_n, v_w \text{ are feminine} \\ \text{man} & \text{if a majority of } v_a, v_c, v_e, v_n, v_w \text{ are masculine} \end{cases}\end{aligned}$$

[9]We underscore that this is merely a toy example. In reality, τ_{gen} would include far more features in its constitutive basis and more genders in its domain, and it would not map the constitutive basis to genders by taking a simple majority.

We can use the alignments τ_{gen} and τ_{audit} to state precisely the quantity of interest in common audit studies. A typical audit study fixes two resumes, Resume_1 and Resume_2, which differ only in their names. Because resumes only contain a few pieces of information about an individual, many different people with different attributes could possess the same resume. Where a "person" refers to a maximally descriptive set of attributes, let $\mathcal{A}_1$ be the set of people with resume Resume_1, and let $\mathcal{A}_2$ be the set of people with resume Resume_2. (In other words, $\mathcal{A}_1$ is the set of people that τ_{audit} aligns with Resume_1, and $\mathcal{A}_2$ is the set of people that τ_{audit} aligns with Resume_2.)

The gender compositions of $\mathcal{A}_1$ and in $\mathcal{A}_2$ will tend to differ, especially given that the names appearing on the resumes Resume_1 and Resume_2, like Andrew and Anne, are highly correlated with gender. Let Gen_1 indicate the gender most common in $\mathcal{A}_1$ and let Gen_2 indicate the gender most common in $\mathcal{A}_2$. Then it follows from the assumption that τ_{gen} and τ_{audit} are causally consistent that audit studies successfully test the causal effect of gender on interview outcomes, in the following sense: the difference in interview status that results from changing Gen_1 to Gen_2 is the same as the difference that results from changing $\mathcal{A}_1$ to $\mathcal{A}_2$, which is in turn the same as changing Resume_1 and Resume_2. Modeling the intervention as on resumes rather than gender allows experimenters to hold fixed the attributes on the resume that are *not* abstracted into gender (for example, an applicant's telephone number). The necessary assumptions of causal consistency are licensed by the social constructionist theory of gender that defined τ_{gen}, and by construction of audit studies, which omit all attributes not on the resume via τ_{audit} (Figure 3).

Formally, audit studies purport to test the effect of gender on interview status via the following equation:

$$\begin{aligned} & \mathbb{P}[\mathsf{Interview\ Callback} \,|\, \text{do}(\text{Gen}_1)] - \mathbb{P}[\mathsf{Interview\ Callback} \,|\, \text{do}(\text{Gen}_2)] \\ = \; & \mathbb{P}[\mathsf{Interview\ Callback} \,|\, \text{do}(\text{Resume}_1)] - \mathbb{P}[\mathsf{Interview\ Callback} \,|\, \text{do}(\text{Resume}_2)]. \end{aligned}$$

Admittedly, this picture is an over-simplification. There are multiple interventions on all of an individual's attributes which could correspond to a change from Gen_1 to Gen_2, so there is significant under-determination and ambiguity in finding a lower-level intervention to test the effect of this change. But this ambiguity is a common feature of causal inference (Spirtes and Scheines, 2004): in saying that we changed a stimulus for the bird from red to blue, it is ambiguous whether we changed it from crimson to cyan, or from scarlet to turquoise. Causal consistency guarantees that all such changes produce the same pattern of changes in outcome. Thus, to the extent that we have causal consistency of τ_{gen} in a diagram like Figure 3, there is a clear and principled sense in which audit studies can be modeled as testing the effect of interventions on gender. More generally, one can model claims about causal discrimination as claims about the effects of gender, understood as an abstraction of lower-level, in-principle manipulable attributes.

The abstraction framework thus addresses the objection that audit studies cannot, in principle, be modeled causally because of constitutive relationships between protected attributes and the attributes constituting them (Kohler-Hausmann, 2018). However, in the next section we use the framework of causal abstraction to distinguish some further challenges for modeling audit studies.

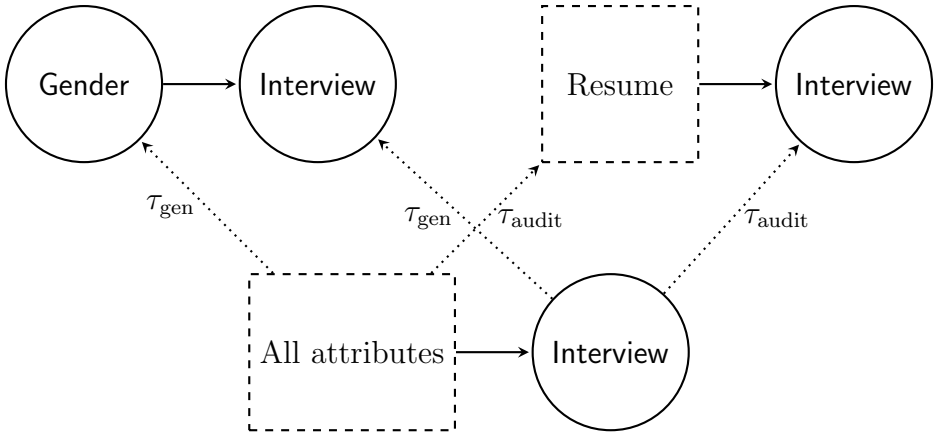


Figure 3: The abstraction model of audit studies. The low-level causal model contains all independently manipulable, low-level attributes that affect interview callbacks. The high level has two separate causal models, requiring two alignments. The first, τ_{audit}, maps low-level attributes to those that appear on the resume. The second, τ_{gen}, represents how gender is constituted by low-level attributes, aligning to a high level model which captures gender's effect on hiring. Audits construct two resumes that differ in gender-coded names and then measure their effect on outcomes; if both alignments are causally consistent, the effect of gender is given by interventions on resumes.

4.2 Challenges for audit studies

We can now use the framework of causal abstraction to distinguish three challenges to the claim that by intervening on names, audit studies reveal a causal effect of a protected attribute that constitutes discrimination:

- *Alignment and high-level model definition (§4.2.1).* There must exist a defensible alignment between gender and its lower-level constituents, including names, to a high level model, where interventions on gender-coded names align to interventions on gender.

- *Assessing causal consistency (§4.2.2).*On this alignment, testing effects of gender-coded names must provide information about the overall effect of gender. At the extreme, this could be achieved if the alignment were perfectly causally consistent, since in that case, each intervention in the low-level model which corresponds (via the alignment) to a change in gender would have the same effect.

- *Normative relevance (§4.2.3).* The interventional quantities revealed by audit studies must correspond to gender subpopulations which are normatively relevant. For instance, a gender subpopulation might be relevant because it is in some sense "typical," or because it has other features of normative relevance to expectations of similar treatment. (To reveal the *absence* of a causal effect of gender, the interventional quantities revealed by audit studies would have to correspond to *all* the gender subpopulations that are normatively relevant.[10])

[10]Many additional assumptions are needed to move from an audit's failure to confirm a causal effect of race or gender to a conclusion that there is not discrimination. Narayanan (2022) outlines many obstacles to detecting these effects quantitatively, such as lacking statistical power (pp. 9-10) or studying the wrong

4.2.1 Defining the alignment and high-level model

Consider, first, the challenge of defining the alignment τ_{gen} to capture a social constructionist theory of gender. Both empirical and normative considerations bear on the choice of alignment, and there are a number of approaches one might take to constructing it. One option is to turn to experts, drawing on leading accounts of gender to determine what features are gendered and how those align to gender statuses. Another option is to learn from data how gender is socially constructed, using techniques such as causal feature learning or ideas from causal discovery, as developed by current work in the social sciences [citation redacted]. A third option would be to poll individuals on their views of gendered attributes and aggregate these opinions. However the alignment is decided, to vindicate the audit study on names, changes in gender-coded names must align to changes in gender.

We note that to capture some theories of gender, additional development of the alignment formalism might be necessary. In our toy example, we align resumes to the most common gender in the population for simplicity, but the alignment could be extended to align resumes to genders in other ways. For example, one could probabilistically align Resume_1 to Gen_1 and Gen_2 according to the frequencies of each in the population $\mathcal{A}_1$ (and handle Resume_2 analogously) to capture indeterminacy. Similarly, one might associate high-level manipulations with stochastic policies of low-level manipulations rather than hard interventions.[11] Additionally, one could model disagreement about constitutive relationships with sets of candidate alignments.[12]

In addition to defining the alignment, there is a second challenge of defining the high-level model. Even if the changes to the names on resumes performed in audit studies correspond via τ_{gen} to changes in gender, an alignment capturing a social constructionist theory of gender will likely give rise to many other ways of implementing interventions on gender. For example, because gendered educational institutions are part of the social meaning of gender, changing the university on a resume from Barnard to Columbia could also change the gender associated with a given resume, and may well have a different effect on interview status than changing the name from "Anne" to "Andrew" (Figure 4). This ambiguity does not arise if the alignment τ_{gen} is perfectly causally consistent, for then the effects of gender correspond perfectly with those on its aligned constituents. However, once we relax the stringent requirement of causal consistency (discussed next in §4.2.2), and allow that the effect of gender on interview status is some complex function of the varying effects of many different lower-level changes, we face the difficult task of aggregating them into a single summary of gender's effect on hiring outcomes. That is, we must define the equation that underlies the arrow $\mathsf{Gender} \to \mathsf{Interview}$ in Figure 3.

Obvious ways of aggregating low-level into high-level effects, such as taking means over distributions of resumes, reflect a normative assumption that theories of gender ought to capture the average experience. One of intersectional theory's founding texts—Kimberlé Crenshaw's (1989) critique of causal legal definitions of discrimination—questions this focus

stage of the hiring process (pp. 13-15). However, these practical challenges are not problems for the very notion of causal discrimination.

[11] This may be necessary in order to ensure that high-level manipulations qualify as interventions on certain theories of causality. See fn. 3 and (Stern and Eva, 2023, fn. 37).

[12] We thank an anonymous reviewer for suggesting this possibility.



(a) Coarse-grained (b) Fine-grained

Figure 4: Measuring how an outcome could be different when moving from one category to another may depend on how one picks exemplars of the two categories. Given a two-dimensional feature space (given by x and y coordinates), suppose settings of those features are clustered into categories, as boxes in Subfigure (a). Shade of teal denotes likelihood of an outcome (say, interview status) for a given category. Subfigure (b) reveals a finer-grained picture of the space of individuals, at which level outcome status is binary (yes or no). Taking an individual from one category to another can be done in any number of different ways, and these different possibilities may also differ with respect to the outcome. To the extent that the alignment is causally consistent, this type of ambiguity will be absent.

on averages, suggesting that we instead look to the experiences of especially marginalized women, namely Black women, to understand gender discrimination. Consequently, aggregation functions informed by intersectional critiques may not use simple averages. For example, one might say that the effect of a high-level intervention that changes gender from man to woman is the *geometric mean* (or the *maximum*) of the effects of all low-level interventions that realize this change in gender.

4.2.2 Causal consistency

After one defines the alignment τ_{gen} and high-level model, there is a further challenge of measuring the alignment's causal consistency.[13] First off, the choice of causal consistency measure, like the high-level model, will involve normative choices about norms of aggregation. To assess the consistency of the alignment, one must aggregate measures of consistency for every possible low-level causal contrast aligning and its corresponding high-level model intervention (as in Definition 4) into a single summary. In this summary, it may not be appropriate to use a simple population average, for the same reasons simple averages may not be appropriate to define the high-level effect; some causal contrasts may be weighted more heavily in a measure of causal consistency than others, reflecting a concern for describing effects that differ from the simple average.

Even once a causal consistency measure is fixed, deciding whether an alignment is causally consistent "enough" is also both a normative and empirical judgment. To obtain a theory of gender that is explanatory across a wider variety of contexts, or one that employs fewer high-level categories, we might have to relax our definition of approximate causal consistency

[13]For a related discussion of challenges to establishing causal consistency of social constructions in a somewhat different context, the application of the potential outcomes framework to observational studies, see Malinsky and Bright (2026).

by increasing ϵ (cf. Definition 4)—which may be justified by different standards of consistency for sweeping social explanations than local explanations. Empirically, this hangs on whether gender proves to be an explanatory way of clustering lower-level attributes, once an aggregation function and measure of causal consistency have been chosen.[14] Current work in social science employs the framework developed here to test whether this is the case over real-world datasets [citation redacted].

4.2.3 Normative relevance

Suppose, however, that these problems are addressed, and that we have defined the effects of gender as some complex function of the effects of its lower-level constituents. Then if the alignment is approximately causally consistent, we could rest assured that audit studies revealed an effect of gender, i.e. that we acquire information about (1) via (2):

$$\mathbb{P}[\mathsf{Interview\ Callback} \mid \mathrm{do}(\mathrm{Gen}_1)] - \mathbb{P}[\mathsf{Interview\ Callback} \mid \mathrm{do}(\mathrm{Gen}_2)] \tag{1}$$

$$\mathbb{P}[\mathsf{Interview\ Callback} \mid \mathrm{do}(\mathrm{Resume}_1)] - \mathbb{P}[\mathsf{Interview\ Callback} \mid \mathrm{do}(\mathrm{Resume}_2)]. \tag{2}$$

Even when this assumption is granted, there remains a second question about why the interventional difference (2) is normatively relevant. For instance, the populations corresponding to the resumes used in the audit study may be *atypical*, insofar as there is no guarantee that when one changes "Andrew" to "Anne" on a resume, one moves from a representative population of men to a representative population of women (Figure 5).

If the resumes created by audit studies are highly atypical, it is unclear how they could shed light on a normatively salient population; the interventions performed by audit studies would simply be "strange" (Hu and Kohler-Hausmann, 2020, p. 10).

[14] It is an open question whether a metaphysically plausible alignment between a protected attribute like race and its constituents can deliver cross-contextual causal consistency; if this is impossible, one might conclude that race does not exist. For arguments along these lines, see Tolbert (2024a, p. 1105) and Tolbert (2024b).

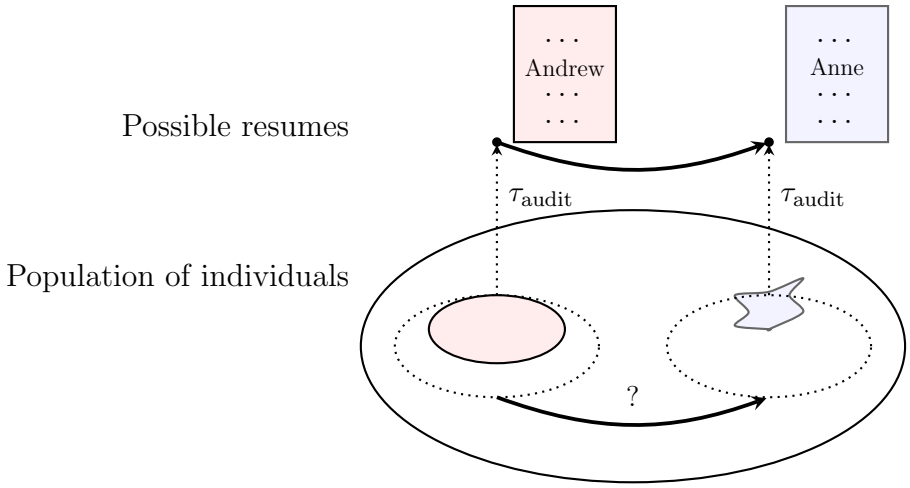


Figure 5: Atypicality in the resume audit study. The resumes differ only in names, but the "Andrew" resume corresponds to a large population of individuals (the red oval), who are representative of the larger group of people named Andrew (the leftmost dotted oval). By contrast, the "Anne" resume corresponds to a small, atypical population of individuals (the blue splotch), rather than the larger group of people named Anne (the rightmost dotted oval).

That said, normative relevance need not require typicality. For example, in the resume audit study, an auditor who believes that women students who study engineering should receive similar job opportunities as men might decide that the causal contrast between women and men engineers is relevant, even if the former population is "atypical" (i.e. significantly smaller). This choice of a causal contrast would be based on a normative assumption that these populations deserve similar treatment. Likewise, concluding from the *absence* of such an interventional difference that there is not discrimination requires a normative assumption that there are no other contrasts that are normatively relevant—for instance, contrasts corresponding to differences in the schools typically attended by women and men.

In sum, a successful model of audit studies of gender discrimination via names would require both an alignment between gender and its constituents (§4.2.1) which conveys information about high-level effects through low-level interventions on names (§4.2.2), and an argument that the populations corresponding to interventions on names are normatively relevant (§4.2.3). The assumption that this can be done may be subjected to further normative and empirical challenges, and we make no claim to have defended it. However, these further debates target particular modeling assumptions, which can be stated within the framework of causal abstraction; unlike the independent manipulability problem, they raise no worries for causal models of discrimination as such. For example, although the obstacle to modeling audit studies is not independent manipulability, it could be a failure of causal consistency—and causal consistency is a notion that is defined within the framework of causal abstraction.

4.3 The expressive power of causal abstraction

We now illustrate how the abstraction framework can be used to capture and clarify disagreements about whether discrimination occurred in a landmark legal case, as well as dis-

agreements about how best to design experiments to test for discrimination.

Legal arguments. The abstraction framework captures disagreements about whether gender discrimination occurred. In particular, the alignment τ_{gen} can represent underlying disagreements about what attributes constitute gender. *Bostock v. Clayton* provides one example. In this case, the plaintiff, a gay male employee, was fired for his sexual orientation. The majority decided that this amounted to sexual discrimination, since had the male employee been a female employee attracted to men, the employee would not have been fired. The dissenters replied that had the male employee been a female employee with same-sex attraction, she would have still been fired (Dembroff et al., 2020; Kohler-Hausmann and Dembroff, 2022). Should we vary the individual's sexual orientation to test for gender discrimination, or just vary biological sex features and hold orientation fixed?[15]

The latter approach, which varies the individual's sex but not their sexual orientation, does not consider sexual orientation as an aspect of gender, and indeed the dissents in SCOTUS (2020) are motivated by the view that effects of sexual orientation do not constitute effects of gender. Thus this view would be represented by an alignment τ_{gen} which does not map sexual orientation to gender. By contrast, to model the majority's view that we can vary sexual orientation to test for gender discrimination, the high-level model of gender in the abstraction could possess a single gender variable with four possible values: gender-conforming man, gender-conforming woman, gender-nonconforming man, and gender-nonconforming woman, with heterosexual orientation aligning via τ_{gen} to gender conformity. The lower-level attribute of the plaintiff being same-sex attracted would then be aligned with being a gender-nonconforming man, so varying sexual orientation is a way of intervening on gender (from gender-conforming man to gender-nonconforming man). Many social constructionist theories of gender indeed intend to highlight norms punishing non-conformity (cf. Kohler-Hausmann and Dembroff 2022, pp. 88-89). Such modeling choices might be further justified, in part, by considerations of causal consistency; if gender-conforming and gender-nonconforming men are in fact treated very differently, then without such a high-level distinction, the abstraction will not be very causally consistent, as it combines groups that are treated very differently.

Experimental design. The abstraction framework also clarifies disagreements about experimental design that are obscured by the Single-Level Assumption. Imagine an in-person audit study, in which trained actors of different genders attend a job interview, presenting identical resumes and answering interview questions identically (Hu 2024a p. 8; cf. Kohler-Hausmann 2018, p. 1216). An experimenter must determine whether to match or experimentally vary the appearances and mannerisms of actors of different genders—for instance, whether they are wearing a dress or a suit, whether or not they wear makeup, or how assertive they are.

The Single-Level Assumption leads to an implausible account of the diversity of experimental designs for hiring audits. When audit studies are modeled using the Single-Level

[15]To clarify, we are not offering a legal interpretation of the SCOTUS opinion on its own terms, but rather showing that our model provides a principled way to distinguish two counterfactual tests of gender discrimination, one similar to that of Dembroff et al. (2020) and another given by the dissent in SCOTUS (2020).

Assumption, experimental designs must always be licensed by bespoke assumptions about independent manipulability. Any features of individuals that experimenters vary are treated as "part of" gender, while any feature they keep constant is treated as "separate from" gender, and placed in the causal diagram subject to an assumption of independent manipulability. When a variable is manipulable independently of gender, it cannot partly constitute gender. Thus, different choices of intervention correspond to different assumptions about how gender is socially constructed. This leads to an implausible proliferation of views about gender: Two auditors with slightly different experimental manipulations (e.g. attire alone vs. attire and cosmetics) must be viewed as having different constitutive bases for gender, even if their experiments appear motivated by similar conceptions of gender.

We solve this puzzle by separating disagreements about the constitutive basis from the choice of causal contrast. In the abstraction framework, experimenters can agree that the constitutive basis includes both attire and cosmetics, but still decide to explore different causal contrasts based on disagreements about normative relevance or causal consistency. For example, one auditor might hypothesize that gender is very causally consistent and manipulating attire is sufficient to represent the effect of all other contrasts, while another might think that gender is more causally inconsistent and postulate manipulating both attire and cosmetics is a more representative empirical probe of gender's overall effect.

The single-level view's tight connection between assumptions about the social construction of gender and an audit study's choice of an intervention opens another puzzle: It can lead to normatively undesirable conclusions about the constitution of gender, even by an auditor's own lights. Suppose an auditor designs a causal test of gender discrimination looking at differences in treatment between gender-nonconforming men and gender-conforming women. To vary gender-conformity across applicants, both applicants wear skirts. This implies, on the Single-level Assumption, that dress is manipulable independently of gender. But this assumption is strange in light of an underlying motivation of the experiment, namely that there are highly gendered standards about dress, which would suggest that dress partially constitutes gender.

By contrast, on the abstraction view, this auditor's design can be viewed as selecting normatively relevant contrast in the absence of perfect causal consistency. The auditor includes dress in their abstraction, corresponding to their understanding that standards of dress are gendered. They then select an intervention that compares women wearing skirts and men wearing skirts, corresponding to the normatively relevant comparison of treatment for gender-conforming women and gender-nonconforming men. Their choice of intervention is made *in light* of how dress constitutes gender, rather than by excluding dress from the constitution of gender.

5 Two notions of causal discrimination

We argued that the abstraction framework for modeling discrimination addresses the independent manipulability problem, and that it provides the precision needed to distinguish several challenges for intervention identification. Some critics of causal approaches to discrimination have used these challenges to argue that discrimination claims involve social-level, rather than individual-level, explanation (Dembroff et al. 2020, p.9). In this section, we show

how causal abstraction can be used to raise an important distinction in such a claim.

Suppose someone explains why a child, Oona, is not allowed to watch Game of Thrones by appealing to her membership in that social category—"because she is a child" (Dembroff et al. 2020, p.9). Dembroff et al. (2020) point out that this sort of explanation operates "at the level of social meanings and norms attached to the category *child*, rather than at the individual level of Oona's age." Yet there is an ambiguity in this observation, because there are two causal contrasts by which social norms can perform explanatory work.

One kind of causal contrast is between two states of a socially embedded category. We might ask: Would Oona have been allowed to watch Game of Thrones were she an adult, rather than a child? To pose this counterfactual, we cannot just vary Oona's age—leaving all else constant—but also features about Oona's social relations that would change with her age, like the way Oona's parents interact with her. In the abstraction framework, we select a low-level intervention on Oona's features that corresponds to a high-level intervention from "child" to "adult," and then measure the causal effect of that intervention. Call this *explanation by a socially constructed attribute.*

A second, relevant causal contrast concerns what the effect on the outcome would have been if the socially constructed category *itself* were constructed differently. We might ask: Would Oona have been allowed to watch Game of Thrones, if the social norms attending to childhood were different? On the abstraction framework, this causal contrast can be represented by changing the abstraction of the social category itself, so that parental supervision of media consumption is *not* partially constitutive of childhood; then, we measure the causal effect of this *constitutive counterfactual* on the outcome of interest, in this case Oona's ability to watch Game of Thrones. Call this *explanation by a social kind's constitutive norms.*

These two explanations answer different causal questions. The first assesses the effect of a social kind *as it is currently defined in our social structure*, while the second identifies the effect of a social kind *if we reformed the social norms defining that kind.* These causal questions correspond to different ways of spelling out the notion of causal discrimination:

> *Attribute Discrimination:* A person is directly discriminated against only if she is treated worse than others, and this is caused by her protected attribute, in the sense that were that attribute, and only that attribute, different, she would not have been treated worse than others.

> *Norm Discrimination:* A person is directly discriminated against only if she is treated worse than others, and this is caused by her protected attribute, in the sense that were that attribute differently constituted, she would not have been treated worse than others.

These should both be seen as live options. The idea of attribute discrimination is articulated in many of the activist-led anti-discrimination efforts in the 1950s that inspired modern anti-discrimination law (Cherry and Bendick Jr., 2018). Likewise, the notion of norm discrimination underlies a prominent theory of algorithmic discrimination by Barocas et al. (2023), who propose an individual is discriminated against if they would have had different opportunities absent structural inequality (cf. Liu and Cohen 2024), and Kohler-Hausmann (2018, p.1226) notes multiple legal theorists have proposed interpretations of anti-discrimination law with this view.

This paper will not adjudicate between these notions of discrimination. However, both are *causal* notions of discrimination, both are amenable to social construction, and both can be formalized using causal abstraction.

6 Conclusion

While pervasive, the notion of causal discrimination is difficult to spell out in a precise and plausible way. Independent manipulability requires that protected attributes like gender and race are separable from other attributes, and social constructionism casts doubt on this requirement. In response, many have proposed that we give up on modeling gender as causing worse treatment, either arguing that a distinct attribute (e.g. perception of gender) causes worse treatment, or that traditional causal models of discrimination are fundamentally misguided.

This paper has addressed the problem differently. We introduced a framework for reasoning about discrimination, in which gender is a high-level abstraction of lower-level, in-principle manipulable features. In this framework, gender can be modeled as itself causing worse treatment. The essential condition of independent manipulability is ensured by allowing assumptions about social construction to be precisely and explicitly stated, via an alignment between gender and its lower-level constituents. Such assumptions can then be subjected to further challenges. How do we define social kinds and their effects? Which population contrasts are normatively relevant to discrimination? These questions have an empirical aspect and a normative one: they recommend different experimental designs, and their answers will depend in part on what makes discrimination wrongful. While answers to these questions lie beyond the scope of this paper, we have used the abstraction framework to state them precisely, and to lay out some of their implications. We conclude that these questions point to important avenues for further work at the intersection between ethics and philosophy of science, but not to any in-principle objections to causal models of discrimination as such.

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